Peer Response
by Mansour Saeed Mubarak Hamdan Al Hamdani - Wednesday, 3 September 2025, 5:53 PM
Your analysis of Agent Communication Languages (ACLs) like KQML provides a solid foundation for understanding their role in multi-agent systems. I agree that the standardized, high-level protocol of ACLs, as noted by Finin et al. (1992), is a key advantage, enabling interoperability across heterogeneous platforms—a critical feature for distributed intelligent systems. The ability to support complex interactions like negotiation and task delegation, as highlighted by Hai-long and Tie-jun (2001), indeed sets ACLs apart, aligning with contemporary AI research in knowledge representation and speech act theory (Chalupsky et al., 1992).
However, your point about the efficiency trade-off is well-taken. The computational overhead of parsing ACLs, especially without a shared ontology (Hendler and Saltz, 1995), can be a significant drawback. This contrasts sharply with the direct, efficient method invocation in Python or Java, which excels in controlled, intra-language settings (Luo et al., 2012). Your conclusion that the choice between ACLs and method invocation hinges on the problem context—multi-agent coordination versus software integration—is insightful and supported by Drasko and Rakic (2024).
To deepen this, consider exploring how recent advancements in ontologies might address semantic interoperability challenges in ACLs. Great work—looking forward to further discussion.
References
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In reply to Abdulla Husain Salem Hadna Almessabi
Peer Response
by Ali Alzahmi - Saturday, 6 September 2025, 7:45 AM
Abdulla, your post gives an excellent introduction to the role of Agent Communication Languages (ACLs), including KQML as a high-level communication protocol in multi-agent systems. I also concur with your argument that their lack of dependence on implementation languages is particularly what makes ACLs useful in heterogeneous and distributed environments, where the agents must be able to coordinate with each other despite being implemented on different platforms. This aligns with recent studies demonstrating how structured communication is vital with larger language model-based systems taking a central role in intelligent applications (Fan et al., 2024).
I also enjoy how you touch upon the fact that ACLs can accommodate more interactive tasks, such as negotiation, cooperation, and delegation of tasks. These upper-level performatives are more important to achieve intelligent agents that are more than data exchange machines since they facilitate joint work and mutual decision-making. This view is further supported by modern research on large language models, which states that communication acts are the source of higher-order reasoning and coordination in multi-agent systems (Su et al., 2024).
Simultaneously, you also made the weaknesses of ACLs quite visible. They are expressive but inefficient; without consensus on ontologies, it is hard to have real semantic interoperability. This corresponds to the results that language models and communication protocols provide flexibility and abstraction, but, at the same time, may also raise the cost of computation and generate ambiguity unless designed well (Hao et al., 2022). It was also appropriate when you compared it to method invocation in Python or Java, demonstrating that, although method calls offer a direct and efficient interaction in tightly-coupled systems, they cannot possibly be as open and interoperable as ACLs are in a distributed system.
Overall, your post provides a good balance of the strong and the weak in ACLs. You demonstrate that the choice of using ACLs or method invocation is mainly determined by the system's emphasis on heterogeneity, flexibility, efficiency, and simplicity.
References:
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In reply to Abdulla Husain Salem Hadna Almessabi
Peer Response
by Ali Alhammadi - Monday, 8 September 2025, 7:21 PM

Abdulla, In your post, there is a good description of how Agent Communication Languages (ACLs), including KQML, are not restricted to mere transfer of basic data but also to convey intentions, goals, and commitments in multi-agent systems (MAS). I also concur that it is the presence of this emphasis on semantics and not just syntax that enables agents to converse meaningfully in open and distributed environments. The same principle is emphasized by the recent studies of large language models as strategic reasoners, where it is pointed out that they must be capable of expressing and comprehending intentions to collaborate on complicated tasks effectively (Zhang et al., 2024).

Your discussion on standardization was also critical to me. As you have stated, ACLs can offer a shared protocol, so that agents written in other languages or operating on other platforms can communicate. This system integration capability is all the more important as AI becomes less about a rule-based agent and more of a language-model-based architecture requiring a trustworthy communications infrastructure in a diverse environment (Huang, 2024). ACLs facilitate scalability and interoperability with a common standard, which is essential in contemporary MAS.

Meanwhile, you are right regarding the limitations analysis. The richness of the semantics of ACLs may impose a computational burden and difficulties with ontological consistency. This can be mirrored in literature on vision-and-language navigation models, where cross-agent differences in interpretation tend to add uncertainty and lower task performance in the real world (Gu et al., 2022). Lightweight alternatives like method invocation in Python or Java may be more efficient in tightly coupled systems, as you rightly note. However, such alternatives may not have the flexibility that ACLs provide in open environments.

Your post, overall, fairly presents both the pros and cons of ACLs, explaining why they are still useful in research regardless of the difficulties of implementing them in large-scale industry.

References:

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